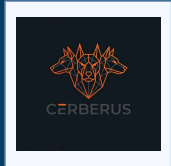


Advance/Decline Breadth Failure



INTERNAL

Detect breadth divergence and leadership narrowing signals in large-cap equity universe.

Author: Cerberus Research

Generated: 2026-08-03 22:46 UTC

CLASSIFICATION

internal

AUTHOR

Cerberus Research

RUN ID

20260803_234626

GIT COMMIT

bccca1c0

DURATION

1.9s

DATASETS

ohlc_v_daily

PARAMETER · INDEX

SPY

PARAMETER · EQUAL_WEIGHT

RSP

PARAMETER · PERIOD

10y

PARAMETER · MA_LENGTH

50

PARAMETER · HORIZON

10

PARAMETER · BASKET

['AAPL', 'MSFT', 'NVDA', 'AMZN', 'GOOGL', 'META', 'TSLA', 'BRK-B', 'JPM', 'XOM', 'LLY', 'UNH', 'V', 'MA', 'COST', 'AVGO', 'HD', 'PG', 'JNJ', 'MRK', 'ABBV', 'PEP', 'KO', 'ADBE', 'NFLX', 'AMD', 'CSCO', 'CRM', 'BAC', 'WMT']

PARAMETER · NO_CHART

False

PARAMETER · SHOW_CHART

False

KEY FINDINGS

- Found 1083 breadth divergence events for SPY

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Generated by the Cerberus studies framework · schema 2

Data Sources

- **ohlcv_daily**: Daily OHLCV for index and breadth basket

Key Findings

- Found 1083 breadth divergence events for SPY

Executive Summary

NOTE

Breadth failure and leadership narrowing study.

Results & Interpretation

NOTE

Detect breadth divergence and leadership narrowing signals in large-cap equity universe.

Main takeaways:

- Found 1083 breadth divergence events for SPY

Objectives

NOTE

Evaluate whether the studied signal produces a measurable, repeatable edge.

Quantify the reliability of the observed patterns across the sample.

Methodology

NOTE

Rolling breadth z-score divergence with equal-weight vs cap-weight comparison.

Charts

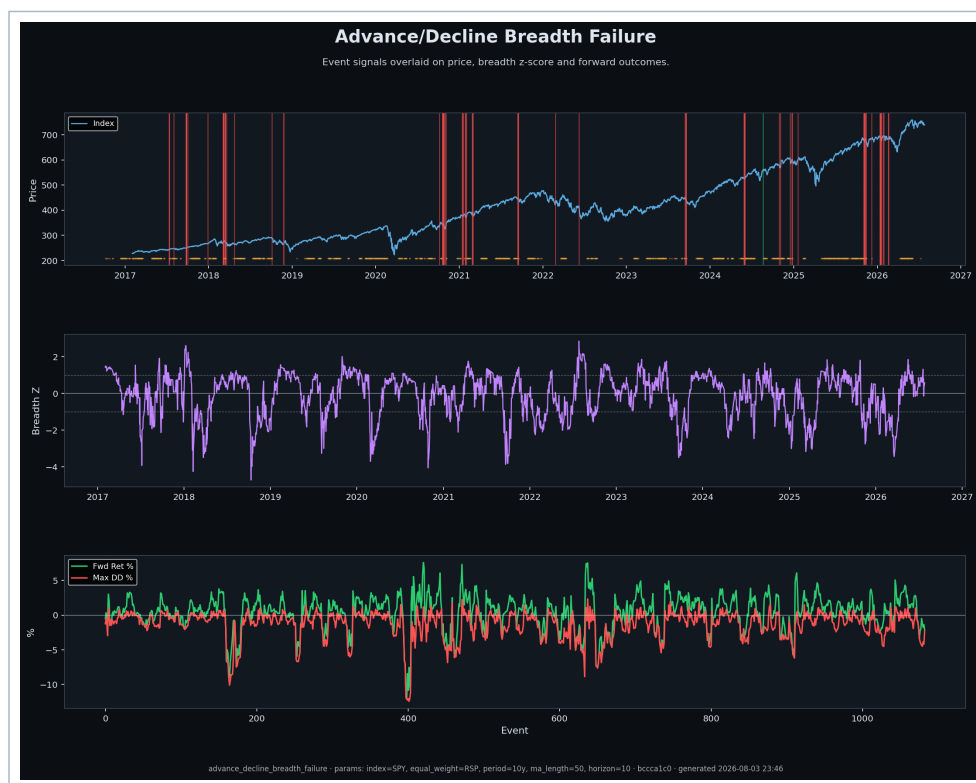


Figure 1. Breadth Failure Events
Breadth Failure Events

Statistical Results

time	breadth_fail...	eqw_diverge...	thrust_exhaust...	fwd_ret_...	max_dd_pct	vol_expansion
2016-10-07	False	True	False	-0.49%	-1.24%	False
2016-10-11	False	True	False	+ 0.35%	-0.49%	False
2016-10-18	False	True	False	-1.26%	-1.26%	False
2016-10-24	False	True	False	-0.81%	-2.95%	False
2016-11-08	False	True	False	+ 3.02%	+ 1.06%	False
2016-11-09	False	True	False	+ 2.00%	+ 0.02%	False
2016-12-13	False	True	False	-1.48%	-1.48%	False
2016-12-14	False	True	False	-0.68%	-0.68%	False
2016-12-15	False	True	False	-1.45%	-1.45%	False
2016-12-16	False	True	False	+ 0.09%	-0.67%	False
2016-12-19	False	True	False	+ 0.47%	-0.89%	False
2016-12-20	False	True	False	0.00%	-1.27%	False
2016-12-21	False	True	False	+ 0.64%	-0.99%	False
2016-12-22	False	True	False	+ 0.48%	-0.82%	False
2016-12-23	False	True	False	+ 0.33%	-0.97%	False
2016-12-27	False	True	False	+ 0.37%	-1.21%	False
2016-12-28	False	True	False	+ 0.95%	-0.39%	False
2016-12-29	False	True	False	+ 1.20%	-0.37%	False
2016-12-30	False	True	False	+ 1.22%	+ 0.77%	False
2017-01-03	False	True	False	+ 0.67%	+ 0.45%	False
2017-01-04	False	True	False	-0.30%	-0.30%	False

Research only. Not investment advice. Past performance is not indicative of future results.

time	breadth_fail...	eqw_diverge...	thrust_exhaust...	fwd_ret_...	max_dd_pct	vol_expansion
2017-01-05	False	True	False	+ 0.15%	-0.22%	False
2017-01-06	False	True	False	-0.47%	-0.57%	False
2017-01-09	False	True	False	+ 0.50%	-0.24%	False
2017-01-11	False	True	False	+ 0.98%	-0.52%	False
2017-01-18	False	True	False	+ 0.38%	-0.37%	False
2017-01-19	False	True	False	+ 0.82%	+ 0.11%	False
2017-01-20	False	True	False	+ 1.15%	-0.26%	False
2017-01-23	False	True	False	+ 1.23%	+ 0.61%	False
2017-02-01	False	True	False	+ 3.21%	+ 0.07%	False
2017-02-02	False	True	False	+ 3.05%	+ 0.51%	False
2017-02-03	False	True	False	+ 2.51%	-0.18%	False
2017-02-06	False	True	False	+ 3.30%	0.00%	False
2017-02-07	False	True	False	+ 3.21%	+ 0.13%	False
2017-02-08	False	True	False	+ 3.14%	+ 0.59%	False
2017-02-14	False	True	False	+ 2.60%	+ 0.44%	False
2017-02-15	False	True	False	+ 1.43%	-0.09%	False
2017-02-16	False	True	False	+ 1.58%	+ 0.16%	False
2017-02-17	False	True	False	+ 1.11%	+ 0.51%	False
2017-02-21	False	True	False	+ 0.22%	-0.09%	False

Table 1. Breadth Events

Showing 40 of 1083 rows; remainder deferred to appendix overflow.

Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

results

time	breadth_fail...	eqw_diverge...	thrust_exhaust...	fwd_ret_...	max_dd_pct	vol_expansion
2016-10-07	False	True	False	-0.49%	-1.24%	False
2016-10-11	False	True	False	+ 0.35%	-0.49%	False
2016-10-18	False	True	False	-1.26%	-1.26%	False
2016-10-24	False	True	False	-0.81%	-2.95%	False
2016-11-08	False	True	False	+ 3.02%	+ 1.06%	False
2016-11-09	False	True	False	+ 2.00%	+ 0.02%	False
2016-12-13	False	True	False	-1.48%	-1.48%	False
2016-12-14	False	True	False	-0.68%	-0.68%	False
2016-12-15	False	True	False	-1.45%	-1.45%	False
2016-12-16	False	True	False	+ 0.09%	-0.67%	False
2016-12-19	False	True	False	+ 0.47%	-0.89%	False
2016-12-20	False	True	False	0.00%	-1.27%	False
2016-12-21	False	True	False	+ 0.64%	-0.99%	False
2016-12-22	False	True	False	+ 0.48%	-0.82%	False
2016-12-23	False	True	False	+ 0.33%	-0.97%	False

Statistics Summary

Key	Value
study	advance_decline_breadth_failure
category	market_internals
status	success
parameters	{'index': 'SPY', 'equal_weight': 'RSP', 'period': '10y', 'ma_length': 50, 'horizon': 10, 'basket': ['AAPL', 'MSFT', 'NVDA', 'AMZN', 'GOOGL', 'META', 'TSLA', 'BRK-B', 'JPM', 'XOM', 'LLY', 'UNH', 'V', 'MA', 'COST', 'AVGO', 'HD', 'PG', 'JNJ', 'MRK', 'ABBV', 'PEP', 'KO', 'ADBE', 'NFLX', 'AMD', 'CSCO', 'CRM', 'BAC', 'WMT'], 'no_chart': False, 'show_chart': False}
datasets	['ohlcv_daily']
rows	0
git_commit	bitcca1c05d8a318e0779c4a3cf1e55db4dc14817
executed_at	2026-08-03T22:46:26.583582+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.
α (alpha)	Intercept or active return versus benchmark.
β (beta)	Sensitivity to a reference factor or market.
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
Framework version	0.1.0
Classification	internal
Generated	2026-08-03 22:46 UTC
Author	Cerberus Research
Study slug	advance_decline_breadth_failure
Run ID	20260803_234626
Git commit	bccca1c0
Datasets	ohlcv_daily
Parameters hash	82a30b21c295
Parameter · index	SPY
Parameter · equal_weight	RSP
Parameter · period	10y
Parameter · ma_length	50
Parameter · horizon	10
Parameter · basket	['AAPL', 'MSFT', 'NVDA', 'AMZN', 'GOOGL', 'META', 'TSLA', 'BRK-B', 'JPM', 'XOM', 'LLY', 'UNH', 'V', 'MA', 'COST', 'AVGO', 'HD', 'PG', 'JNJ', 'MRK', 'ABBV', 'PEP', 'KO', 'ADBE', 'NFLX', 'AMD', 'CSCO', 'CRM', 'BAC', 'WMT']
Parameter · no_chart	False
Parameter · show_chart	False

Field	Value
Data fingerprint	de71f8e33cb9
Duration	1.9s